Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

Canatu Finland Oy, Vantaa

Mechanical Design Engineer

Dear Salla Oksala and the Canatu team,

I'd like to apply for the Mechanical Design Engineer role. Designing equipment that turns an experimental idea into something that works reliably is the part of engineering I enjoy most, and doing that for Canatu's CNT process sounds like a great place to grow.

As a research assistant at Aalborg University I designed a hybrid shoulder exoskeleton. A variable stiffness mechanism, with a worm gear that pretensions the spring, works in parallel with a 6 Nm motor. I optimised it with a genetic algorithm (diameter, spring stiffness and pretension force), made the 3D models, assemblies and drawings in SolidWorks, 3D printed the first parts and had the final ones manufactured. I integrated the motor, encoder and force/torque sensors and measured its stiffness on a physical test setup. It uses over 25% less motor energy and won the Best Paper Award at ISRM 2026.

I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM) and estimated its stiffness with less than 10% error, which needed careful mechanical design and precise testing. In my Master's I worked in a team of 4 electronics and mechanical engineers on a soft exosuit, fabricated silicone-based pneumatic muscles and designed the outer PID loop, with integral anti-windup and error nonlinearisation, of the controller in LabVIEW. In my Bachelor's I did the CAD, machining and assembly of an arm exoskeleton and exo glove. I also hold a Senior Diploma in Fine Arts and do graphic design (Illustrator, Photoshop, Procreate), which helps me sketch ideas and make clear drawings.

I would love to bring new and innovative ideas to your equipment and grow with your mechanical team. My portfolio, with videos and results from my work, is at arunabha22.github.io/arunabha-majumder. Thank you for reading.

Best,
Arunabha Majumder